



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - Prova scritta preliminare del 23/11/2022

PROBLEM - We observe N samples of the following discrete-time real-valued w.s.s. random process:

$$X[n] = S[n] + W[n] = S + W[n], \quad n = 0, 1, 2, \dots, N-1,$$

where S is a constant signal modelled as an unknown deterministic parameter and $W[n]$ is additive correlated Gaussian noise (AGN), zero mean, with power σ_w^2 , modelled as an Auto-Regressive process of order 1, i.e. $W[n] = \rho W[n-1] + I[n]$, where ρ is the one-lag correlation coefficient of the noise and $I[n]$ is the white innovation process with power $\sigma_I^2 = (1 - \rho^2) \sigma_w^2$.

(1) Derive the autocorrelation function (ACF) and the Power Spectral Density (PSD) of the observed process $X[n]$.

(2) Derive the covariance matrix of the noise vector $\mathbf{W} = [W[0] \ W[1] \ \dots \ W[N-1]]^T$ and the probability density function (pdf) of the data vector: $\mathbf{X} = [X[0] \ X[1] \ \dots \ X[N-1]]^T = S\mathbf{1} + \mathbf{W}$.

(3) Derive the Maximum Likelihood (ML) estimator of the signal of interest S , given the observation of $\mathbf{X}$ and comment on the result.

(4) Derive the bias $b = E\{S - \hat{S}_{ML}\}$ and the Mean Square Error (MSE) $\sigma_\varepsilon^2 \triangleq E\{(S - \hat{S}_{ML})^2\}$ of the ML estimator; plot the MSE versus ρ and comment the result.

Hint:

$$\mathbf{M} = \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{N-1} \\ \rho & 1 & \rho & \dots & \rho^{N-2} \\ \rho^2 & \rho & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \rho \\ \rho^{N-1} & \rho^{N-2} & \dots & \rho & 1 \end{bmatrix} \Rightarrow \mathbf{M}^{-1} = \frac{1}{1 - \rho^2} \begin{bmatrix} 1 & -\rho & 0 & \dots & 0 & 0 \\ -\rho & 1 + \rho^2 & -\rho & \dots & 0 & 0 \\ 0 & -\rho & 1 + \rho^2 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -\rho & 0 \\ 0 & 0 & \ddots & -\rho & 1 + \rho^2 & -\rho \\ 0 & 0 & \dots & 0 & -\rho & 1 \end{bmatrix}$$



Solution

(1) ACF and PSD of the process $X[n]$.

$$R_X[m] = E\{X[n]X[n+m]\} = E\{(S+W[n])(S+W[n-m])\} = S^2 + R_W[m] = S^2 + \sigma_W^2 \rho^{|m|}$$

$$S_X(e^{j2\pi F}) = FT\{R_X[m]\} = FT\{S^2\} + FT\{R_W[m]\} = 2\pi S^2 \delta(e^{j2\pi F} - 1) + \frac{(1-\rho^2)\sigma_W^2}{1+\rho^2-2\rho\cos(2\pi F)}$$

(2) Covariance matrix of the noise vector $\mathbf{W}$ and pdf of $\mathbf{X}$.

The covariance matrix $\mathbf{C}_W = E\{\mathbf{W}\mathbf{W}^T\}$ is Toeplitz and symmetric and the element (i,j) is given by

$[\mathbf{C}_W]_{i,j} = R_W[i-j] = \sigma_W^2 \rho^{|i-j|}$, $i, j = 1, 2, \dots, N$. Hence, the covariance matrix is as follows:

$$\mathbf{C}_W = \sigma_W^2 \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{N-1} \\ \rho & 1 & \rho & \dots & \rho^{N-2} \\ \rho^2 & \rho & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \rho \\ \rho^{N-1} & \rho^{N-2} & \dots & \rho & 1 \end{bmatrix} = \sigma_W^2 \mathbf{M}$$

where $\mathbf{M}$ is the normalized noise covariance matrix. The pdf of $\mathbf{X}$ can be derived by observing that $X[n]$ is a Gaussian process, so $\mathbf{X}$ is a Gaussian vector. In fact, S is deterministic and $W[n]$ is a Gaussian process, so also $X[n]$ is a Gaussian random process and $\mathbf{X}$ is a Gaussian vector:

$$\mathbf{X} \in N(\boldsymbol{\eta}_X, \mathbf{C}_X), \text{ where } \boldsymbol{\eta}_X = E\{\mathbf{X}\} = E\{S\mathbf{1} + \mathbf{W}\} = S\mathbf{1} \text{ and } \mathbf{C}_X = \mathbf{C}_W = \sigma_W^2 \mathbf{M}.$$

(3) Maximum Likelihood (ML) estimator of S & (4) Bias and MSE of the ML estimator

The pdf of $\mathbf{X}$ is given by:

$$f_X(\mathbf{x}; S) = \frac{1}{(2\pi)^{N/2} \sqrt{|\sigma_W^2 \mathbf{M}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - S\mathbf{1})^T (\sigma_W^2 \mathbf{M})^{-1} (\mathbf{x} - S\mathbf{1})\right)$$



Hence, the log-likelihood function is:

$$\begin{aligned}
 \ln LF(S) &= \ln f_{\mathbf{x}}(\mathbf{x}; S) = k - \frac{1}{2}(\mathbf{x} - S\mathbf{1})^T (\sigma_w^2 \mathbf{M})^{-1} (\mathbf{x} - S\mathbf{1}) \\
 &= k - \frac{1}{2\sigma_w^2} (\mathbf{x} - S\mathbf{1})^T \mathbf{M}^{-1} (\mathbf{x} - S\mathbf{1}) = k - \frac{1}{2\sigma_w^2} (\mathbf{x}^T \mathbf{M}^{-1} \mathbf{x} - \mathbf{x}^T \mathbf{M}^{-1} \mathbf{1} S - \mathbf{1}^T \mathbf{M}^{-1} \mathbf{x} S + \mathbf{1}^T \mathbf{M}^{-1} \mathbf{1} S^2) \\
 &= k - \frac{1}{2\sigma_w^2} (\mathbf{x}^T \mathbf{M}^{-1} \mathbf{x} - 2(\mathbf{x}^T \mathbf{M}^{-1} \mathbf{1}) S + (\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}) S^2) \\
 &= k - \frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}}{2\sigma_w^2} \left(S^2 - 2 \left(\frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{x}}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}} \right) S + \frac{\mathbf{x}^T \mathbf{M}^{-1} \mathbf{x}}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}} \right)
 \end{aligned}$$

$$\frac{d \ln LF(S)}{dS} = -\frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}}{2\sigma_w^2} \left(2S - 2 \left(\frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{x}}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}} \right) \right) = \frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}}{\sigma_w^2} \left(\frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{x}}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}} - S \right) \equiv I(S) (\mathbf{g}(\mathbf{x}) - S)$$

Hence, $\mathbf{g}(\mathbf{x})$ is the ML estimator, it is unbiased and efficient:

$$\hat{S}_{ML} = g(\mathbf{x}) = \frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{x}}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}}, \quad E\{\hat{S}_{ML}\} = S, \quad MSE\{\hat{S}_{ML}\} = CRB(S) = \frac{1}{I(S)} = \frac{\sigma_w^2}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}}$$

$$\hat{S}_{ML} \text{ is linear and so Gaussian distributed: } \hat{S}_{ML} \in N\left(S, \frac{\sigma_w^2}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}}\right).$$

The ML estimate of S is the output of the linear whitening matched filter (WMF), which is the cascade of a whitening filter (WF) and a matched filter (MF). Remember that the MF output is the ML estimate of S in white Gaussian noise.

The ML estimator in correlated Gaussian noise modelled as an AR(1) process can be expressed as follows:

$$\mathbf{M} = \begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{N-1} \\ \rho & 1 & \rho & \cdots & \rho^{N-2} \\ \rho^2 & \rho & 1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \rho \\ \rho^{N-1} & \rho^{N-2} & \cdots & \rho & 1 \end{bmatrix} \Rightarrow \mathbf{M}^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho & 0 & \cdots & 0 & 0 \\ -\rho & 1+\rho^2 & -\rho & \cdots & 0 & 0 \\ 0 & -\rho & 1+\rho^2 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -\rho & 0 \\ 0 & 0 & \ddots & -\rho & 1+\rho^2 & -\rho \\ 0 & 0 & \cdots & 0 & -\rho & 1 \end{bmatrix}$$



$$\mathbf{1}^T \mathbf{M}^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho & 0 & \dots & 0 & 0 \\ -\rho & 1+\rho^2 & -\rho & \dots & 0 & 0 \\ 0 & -\rho & 1+\rho^2 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -\rho & 0 \\ 0 & 0 & \ddots & -\rho & 1+\rho^2 & -\rho \\ 0 & 0 & \dots & 0 & -\rho & 1 \end{bmatrix}$$

$$\mathbf{1}^T \mathbf{M}^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1-\rho & (1-\rho)^2 & \dots & (1-\rho)^2 & 1-\rho \end{bmatrix} = \frac{1}{1+\rho} \begin{bmatrix} 1 & 1-\rho & \dots & 1-\rho & 1 \end{bmatrix}$$

$$\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1} = \frac{1}{1+\rho} \begin{bmatrix} 1 & 1-\rho & \dots & 1-\rho & 1 \end{bmatrix} \mathbf{1} = \frac{1}{1+\rho} [2 + (N-2)(1-\rho)] = \frac{2\rho + N(1-\rho)}{1+\rho}$$

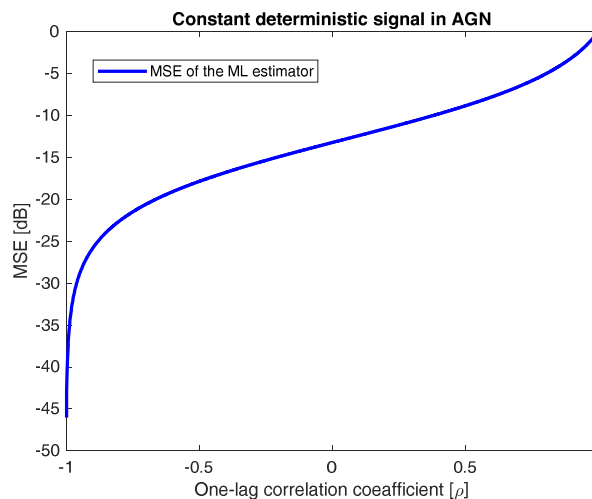
Hence, the ML estimator and its MSE can be expressed as follows:

$$\begin{aligned} \hat{S}_{ML} &= \frac{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{x}}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}} = \frac{1}{2\rho + N(1-\rho)} \begin{bmatrix} 1 & (1-\rho) & \dots & (1-\rho) & 1 \end{bmatrix} \mathbf{x} \\ &= \frac{1}{2\rho + N(1-\rho)} [X[0] + (1-\rho)X[1] + \dots + (1-\rho)X[N-2] + X[N-1]] \end{aligned}$$

$$MSE\{\hat{S}_{ML}\} = \frac{\sigma_w^2}{\mathbf{1}^T \mathbf{M}^{-1} \mathbf{1}} = \frac{(1+\rho)\sigma_w^2}{2\rho + N(1-\rho)}$$

Note that if $\rho = 0$, we have $MSE\{\hat{S}_{ML}\} = \frac{\sigma_w^2}{N}$.

Plot of the MSE ($N = 21, \sigma_w^2 = 1$)





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Note that the useful signal spectrum is a *sinc* centered at $F=0$, where F is the digital frequency. This is the reason why the MSE is minimal ($=0$) when $\rho = -1$ and maximal ($=\sigma_w^2$) when $\rho = 1$.